Exercise Set 4: Topological Invariants

In the previous set we added topology to our space and gained continuity and convergence. But we asked: when are two topological spaces truly the same? And what structure guarantees that limits are unique? In this set we study ***topological invariants*** *properties preserved under homeomorphisms and add the* ***Hausdorff condition****, which gives us unique limits and is the key hypothesis for manifolds later.*

Recall of Definitions and Examples

Homeomorphisms

R1. Write the definition of a **homeomorphism** $f : X \to Y$. What are the three conditions? Why is it not enough for f to be a continuous bijection?

R2. For each of the following pairs, state whether the spaces are homeomorphic. If yes, give an explicit homeomorphism. If no, name a topological invariant that distinguishes them.

(a) $(0, 1)$ and $\mathbb{R}$

(b) $[0, 1]$ and $(0, 1)$

(c) S^1 and $\mathbb{R}$

(d) $[0, 1]$ and $[0, 1]^2$

(e) $\mathbb{R}$ and $\mathbb{R}^2$

R3. Give an example of a continuous bijection $f : X \to Y$ that is *not* a homeomorphism. What goes wrong?

Topological invariants

R4. Write the definition of a **topological invariant**. Give three examples of topological invariants one numerical, one categorical, and one algebraic.

R5. For each of the following properties, decide whether it is a topological invariant. Justify briefly.

(a) Compactness

(b) Having exactly n elements

(c) Being a metric space

(d) Connectedness

(e) Being a subset of $\mathbb{R}^2$

Hausdorff spaces

R6. Write the definition of a **Hausdorff space**. Draw a picture illustrating the condition for two points $x \neq y$.

R7. For each space, determine whether it is Hausdorff.

(a) $\mathbb{R}$ with the standard topology

(b) $X = \{a, b\}$ with topology $\{\emptyset, \{a\}, X\}$

(c) Any discrete topological space

(d) $\mathbb{R}$ with the co-finite topology

(e) Any metric space (X, d)

R8. State the theorem: *in a Hausdorff space, limits of sequences are unique.* Write the proof in your own words. Why does the proof fail in non-Hausdorff spaces?

Compactness and connectedness

R9. Write the definition of **compactness** using open covers. Give two examples of compact spaces and two non-examples, with justification.

R10. Write the definitions of **connected** and **path-connected**. Give an example showing the two are not equivalent.

R11. For each of the following subsets of $\mathbb{R}$, compute the closure $\bar{A}$, the interior $\text{int}(A)$, and the boundary ∂A:

(a) $A = (0, 1)$

(b) $A = [0, 1) \cup \{2\}$

(c) $A = \mathbb{Q}$

(d) $A = \{1/n : n \in \mathbb{N}\}$

Theory

Invariants and homeomorphisms

T1. Prove that homeomorphism is an equivalence relation on the class of topological spaces.

T2. Prove that connectedness is a topological invariant: if $X \cong Y$ and X is connected, then Y is connected.

T3. Use the fact that removing a point from $\mathbb{R}$ disconnects it (but removing a point from $\mathbb{R}^2$ does not) to prove that $\mathbb{R} \ncong \mathbb{R}^2$.
Hint: suppose $f : \mathbb{R} \to \mathbb{R}^2$ is a homeomorphism and consider f restricted to $\mathbb{R} \setminus \{0\}$.

Hausdorff condition

T4. **(Uniqueness of limits)** Prove: in a Hausdorff space, a sequence can converge to at most one point.

T5. Prove: every metric space is Hausdorff.
Hint: given $x \neq y$ with $d(x, y) = r > 0$, use the balls $B_{r/2}(x)$ and $B_{r/2}(y)$.

T6. Prove: in a Hausdorff space, every singleton $\{x\}$ is a closed set.

T7. Prove: every compact subset K of a Hausdorff space X is closed.
Hint: fix $x \notin K$. For each $y \in K$ find disjoint open sets $U_y \ni x$ and $V_y \ni y$. Cover K by finitely many V_y's.

Compactness and continuity

T8. Prove: the continuous image of a compact space is compact.

T9. Prove: the continuous image of a connected space is connected.

T10. **(Extreme Value Theorem)** Let $f : X \to \mathbb{R}$ be continuous with X compact. Prove that f attains its maximum and minimum.
Hint: $f(X)$ is compact in $\mathbb{R}$, hence closed and bounded by Heine–Borel. Then $\sup f(X) \in f(X)$.

T11. **(Closed Map Lemma)** Let $f : X \to Y$ be a continuous bijection with X compact and Y Hausdorff. Prove that f is a homeomorphism.
This is the key lemma used to construct homeomorphisms in practice: instead of verifying continuity of f^{-1}, verify compactness of X and Hausdorffness of Y.

Computation

C1. For each of the following spaces, determine whether it is: (i) Hausdorff, (ii) connected, (iii) compact. Justify each answer.

(a) $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$

(b) $\mathbb{R}^2 \setminus \{(0, 0)\}$

(c) $\mathbb{Q}$ with the subspace topology from $\mathbb{R}$

(d) $[0, 1] \times [0, 1]$ with the product topology

(e) The Cantor set $C \subset [0, 1]$

C2. Use topological invariants to show the following spaces are pairwise non-homeomorphic:

$$\mathbb{R}, \qquad S^1, \qquad [0, 1], \qquad \mathbb{R}^2, \qquad \mathbb{R}^2 \setminus \{0\}$$

Fill in the table with the value of each invariant for each space:

Space	Compact	Connected	Hausdorff	# comp. after removing a point
$\mathbb{R}$				
S^1				
$[0,1]$				
$\mathbb{R}^2$				
$\mathbb{R}^2 \setminus \{0\}$				

C3. Return to the dynamical system $f : X \to X$ from Exercise Sets 2 and 3. Let $X = [0,1]$ with the standard topology, and let $f : X \to X$ be continuous.

(a) Explain why X is compact. Is it Hausdorff?

(b) By the Extreme Value Theorem, f attains its maximum and minimum. Use this to argue that the orbit $\mathcal{O}^+(x_0)$ is always bounded.

(c) Prove that f must have a **fixed point**: there exists $x^* \in X$ such that $f(x^*) = x^*$.
Hint: consider $g(x) = f(x) - x$ and use the Intermediate Value Theorem.

(d) Is the fixed point unique? Give an example where it is not.

(e) Now let $X = (0,1)$ (open interval). Does f still necessarily have a fixed point? Give a counterexample if not.

C4. Let $f : X \to Y$ be a continuous bijection where $X = [0,1)$ and $Y = S^1$ is the circle, defined by $f(t) = (\cos 2\pi t, \sin 2\pi t)$.

(a) Show that f is a continuous bijection.

(b) Show that f is *not* a homeomorphism by finding an invariant that distinguishes $[0,1)$ from S^1.

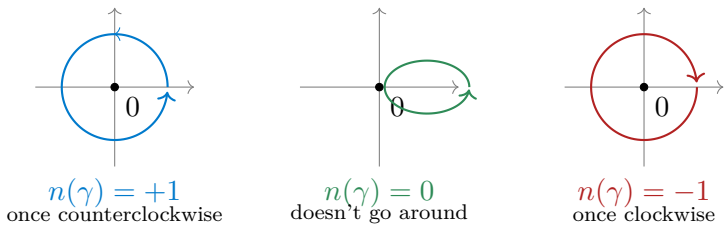
(c) Why does the Closed Map Lemma not apply here?

Physics: The Winding Number Physics Detecting Topology

So far we have been using topology to understand physics. In this section we pause and look at the other direction: ***physics computing topological invariants****. The central idea is that a physical quantity the total charge measured by a field turns out to be a topological invariant of the space the field lives on. It detects holes.*

The winding number

Let $\gamma : [0,1] \to \mathbb{R}^2 \setminus \{0\}$ be a closed curve (so $\gamma(0) = \gamma(1)$) that avoids the origin. Intuitively, the **winding number** $n(\gamma) \in \mathbb{Z}$ counts how many times γ goes around the origin with sign depending on direction.



P1. Computing winding numbers. For each of the following curves in $\mathbb{R}^2 \setminus \{0\}$, determine the winding number $n(\gamma)$ by drawing the curve and counting how many times it goes around the origin.

(a) $\gamma(t) = (\cos 2\pi t, \sin 2\pi t)$ the unit circle, once counterclockwise

(b) $\gamma(t) = (\cos 4\pi t, \sin 4\pi t)$ the unit circle, twice

(c) $\gamma(t) = (\cos 2\pi t + 2, \sin 2\pi t)$ circle centred at $(2, 0)$

(d) $\gamma(t) = (\cos 2\pi t, \sin 2\pi t \cdot (-1))$ circle traversed clockwise

(e) A figure-eight curve whose two loops both enclose the origin

(f) A figure-eight curve whose two loops are on opposite sides of the origin

P2. Winding number is a topological invariant. Two curves γ_0 and γ_1 in $\mathbb{R}^2 \setminus \{0\}$ are called *homotopic* if one can be continuously deformed into the other while staying in $\mathbb{R}^2 \setminus \{0\}$.

(a) Draw two curves in $\mathbb{R}^2 \setminus \{0\}$ with $n = 1$ that look different but are homotopic to each other.

(b) Draw a curve with $n = 1$ and a curve with $n = 0$. Explain intuitively why they are *not* homotopic what would the deformation have to cross?

(c) Conclude: the winding number is constant on homotopy classes. It is a topological invariant of the curve.

P3. The hole makes the difference.

(a) In $\mathbb{R}^2$ (no hole), every closed curve can be continuously shrunk to a point. What does this say about the winding number of every curve in $\mathbb{R}^2$?

(b) In $\mathbb{R}^2 \setminus \{0\}$ (one hole), a curve with $n = 1$ *cannot* be shrunk to a point. Why not?

(c) Conclude: $\mathbb{R}^2$ and $\mathbb{R}^2 \setminus \{0\}$ are **not homeomorphic**. What invariant distinguishes them?

(d) We showed in Theory problem T3 that $\mathbb{R} \ncong \mathbb{R}^2$ using connectedness after removing a point. Use a similar argument involving what happens after removing a point to show $\mathbb{R}^2 \ncong \mathbb{R}^2 \setminus \{0\}$.

P4. Gauss's law physics detecting the hole.

Place a point charge $Q > 0$ at the origin in $\mathbb{R}^2$. The electric field is

$$\mathbf{E}(x, y) = \frac{Q}{2\pi} \cdot \frac{(x, y)}{x^2 + y^2}, \qquad (x, y) \neq (0, 0).$$

For a closed curve γ in $\mathbb{R}^2 \setminus \{0\}$, define the *flux integral*

$$\Phi(\gamma) = \oint_\gamma \mathbf{E} \cdot \hat{n}\, d\ell$$

where $\hat{n}$ is the outward unit normal to γ.

(a) Compute $\Phi(\gamma)$ for $\gamma =$ the unit circle traversed once counterclockwise. Show that $\Phi = Q$.

(b) Now deform the circle into a square surrounding the origin. Argue (without computing) that Φ should give the same answer. Why?
(This is Gauss's law: the flux depends only on whether the curve encloses the charge, not on the shape of the curve.)

(c) If γ does not enclose the origin ($n(\gamma) = 0$), show that $\Phi(\gamma) = 0$.
Hint: on $\mathbb{R}^2 \setminus \{0\}$, away from the charge, the field satisfies $\nabla \cdot \mathbf{E} = 0$.

(d) Conclude: the flux $\Phi(\gamma)$ depends only on the winding number $n(\gamma)$. Specifically, $\Phi(\gamma) = Q \cdot n(\gamma)$. The charge Q is detected by how many times the curve winds around it.

(e) **The key observation.** The charge Q is a property of the *field configuration* on $\mathbb{R}^2 \setminus \{0\}$. If you continuously deform the field (without creating new charges), Q cannot change. **The charge is a topological invariant of the field.**

Can there be a field like $\mathbf{E}$ on all of $\mathbb{R}^2$ (without removing the origin) with the same behaviour? Why or why not?

P5. Topology of the configuration space. The fact that winding numbers are nonzero on $\mathbb{R}^2 \setminus \{0\}$ but always zero on $\mathbb{R}^2$ reflects a fundamental difference in the *topology* of these spaces.

(a) The set of all closed curves in $\mathbb{R}^2 \setminus \{0\}$ (up to homotopy) can be labelled by their winding number. What set do these labels form? What does this tell you about the "shape" of $\mathbb{R}^2 \setminus \{0\}$?

(b) In $\mathbb{R}^2$, how many homotopy classes of closed curves are there? And in $\mathbb{R}^2 \setminus \{0\}$?

(c) **Preview.** The collection of homotopy classes of loops based at a point in a space X forms a group called the *fundamental group* $\pi_1(X)$. Based on your answers:

$$\pi_1(\mathbb{R}^2) = ? \qquad \pi_1(\mathbb{R}^2 \setminus \{0\}) = ?$$

We will study this properly when we reach algebraic topology. For now, the winding number has already given you the answer.

What structure is still missing? We can talk about continuous maps and topological invariants, but not about *smooth* maps, *derivatives*, or *speed*. For that, we need a manifold structure the topic of the next set.***Reflection.*** *In this set we saw how topology gives us invariants properties preserved under homeomorphism that let us distinguish spaces. The Hausdorff condition gave us unique limits, which is the starting point for manifolds. The winding number showed us how physics (electric field, Gauss's law) naturally computes a topological invariant, detecting the hole in* $\mathbb{R}^2 \setminus \{0\}$. *What structure is still missing? We can talk about continuous maps and topological invariants, but not about* smooth *maps,* derivatives*, or* speed*. For that, we need a manifold structure the topic of the next set.*